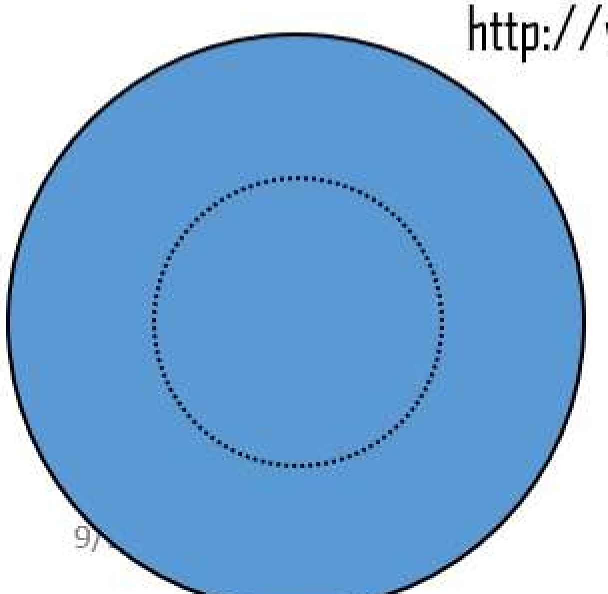
# Welding as a manufacturing process

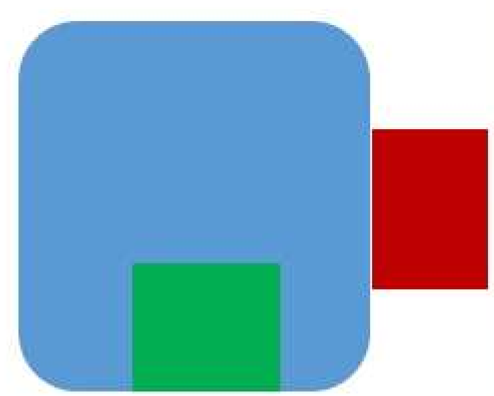
- To produce a product it is often required that two or more parts need to be joined
- Joining can be done by
  - Temporary joining
    - Nut, bolts, screws
  - Semi-permanent joining
    - Tight fits, riveting
  - Permanent joining
    - Welding
- Welding is permanent joining of material
  - Really difficult to "unjoin"; requires machining to unjoin
  - Possible by bonding at the Atomistic lev
  - The joint strength should be at least strength of the base material (Better ex

# Need for welding

- Should be avoided (if possible)but usually not possible to eliminate it completely because of
  - Complex shape of product needed
  - Different functionality (hence properties) of components required
  - Helps in modular design ad fabrication
  - Also needed for fabrication large part eg tankers, cross country pipelines, turbines etc.

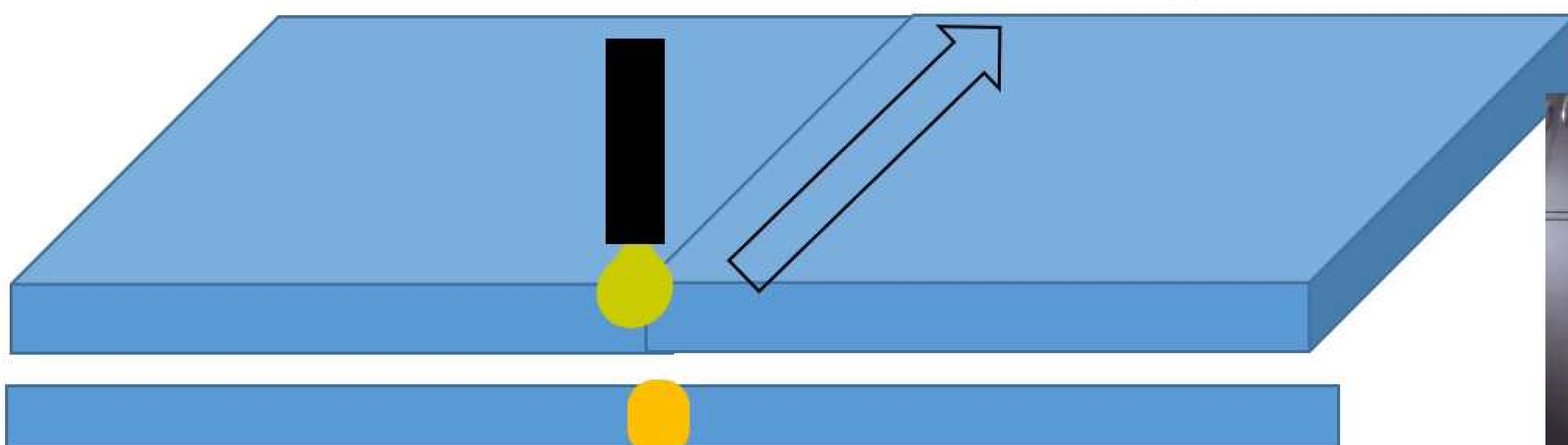

http://www.aukevisser.nl/supertankers/id23.htm





# Welding mechanisms

- Welding without melting (Solid state welding)
  - Clean the material of contamination
    - before bringing the surfaces in contact
  - Increase the area of contact
    - Making the surface very smooth
  - Apply pressure/force to increase the area of contact
    - Apply heat to lower the requirement of force
- Welding by melting of materials (fusion welding
  - Keep the surfaces (to be joined) together
  - Then melt both the surfaces together



# Types of welding processes

- Solid state welding processes

  Forge welding
  Friction welding
  Friction stir welding
  Diffusion bonding
  Ultrasonic welding
  Explosive welding

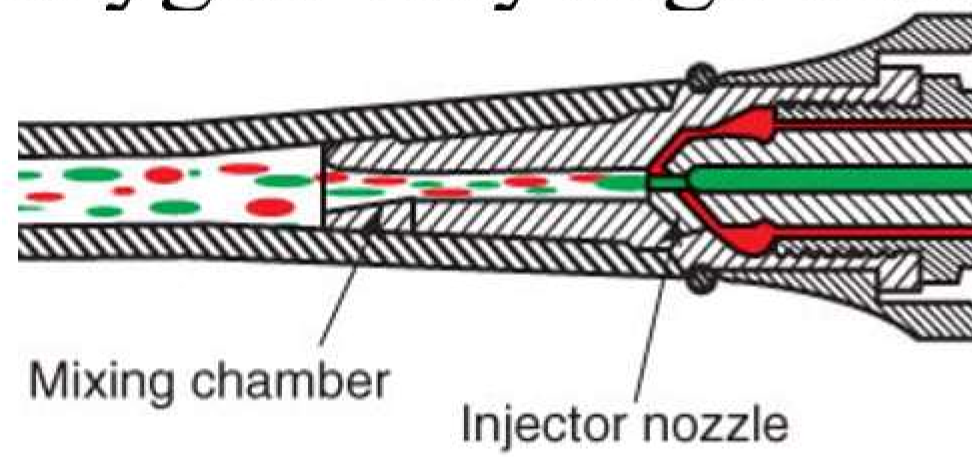
- Fusion welding processes(Based on types of heat sources
  - Chemical energy
    - Gas welding
    - Thermit welding
  - Electrical Energy
    - Arc welding
    - Resistance welding
  - High Energy beam
    - Laser Beam welding
    - Electron Beam welding

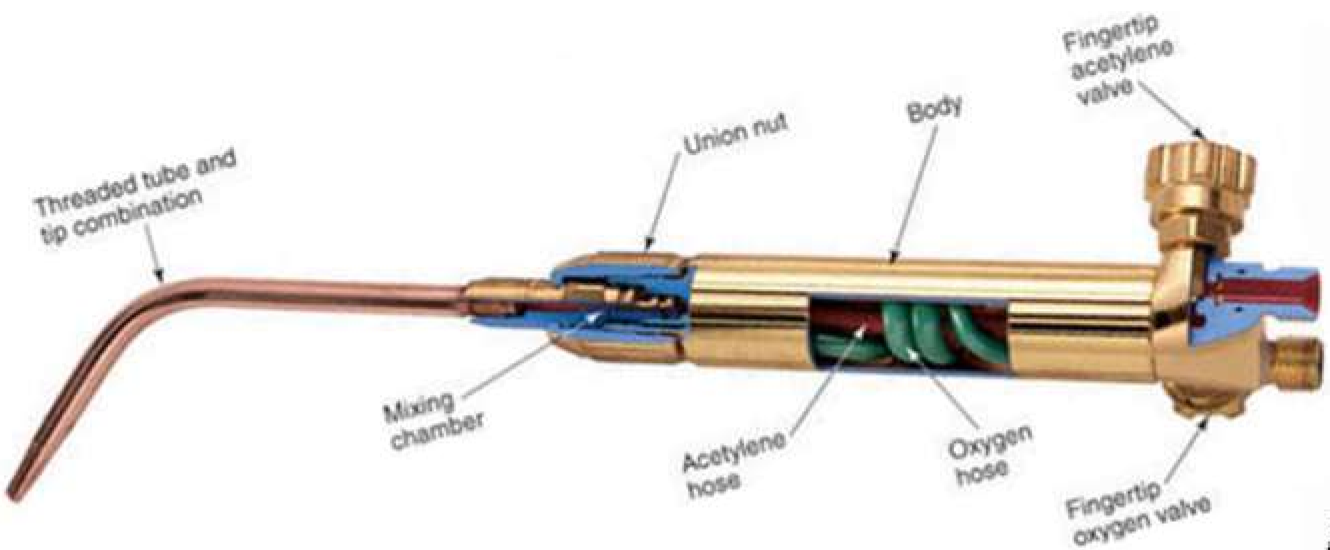
- Other basis of classification
  - Using filler material
  - Without using filler material (Autogenous welding)

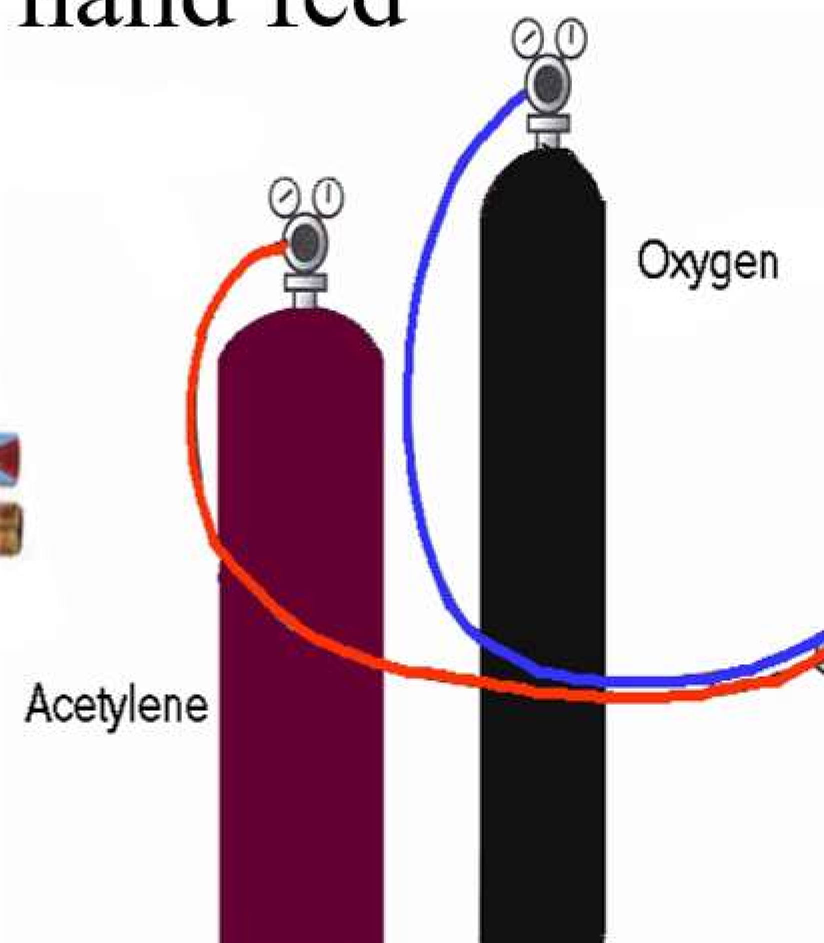
- Welding allied processes
  - brazing and soldering
  - Gas cutting, arc cutting and plasma cutting

# Gas Welding (Oxy-Fuel welding)

- Source of heat is burning fuel
- Burning gaseous fuel premixed with air higher temperature
  - Higher rate of reaction; more max temperature
- Burning fuel gas premixed with pure oxygen very high temp
  - Around 3000°C, easily melts all metals
- Any fuel gas can be used
  - LPG, butane, acetylene, hydrogen etc.
- Filler material, if needed is hand fed
  - Also melted in the flame




http://www.halversoncts.com/116-oxyacetylene-torch-types.html,

# Gas Welding setup

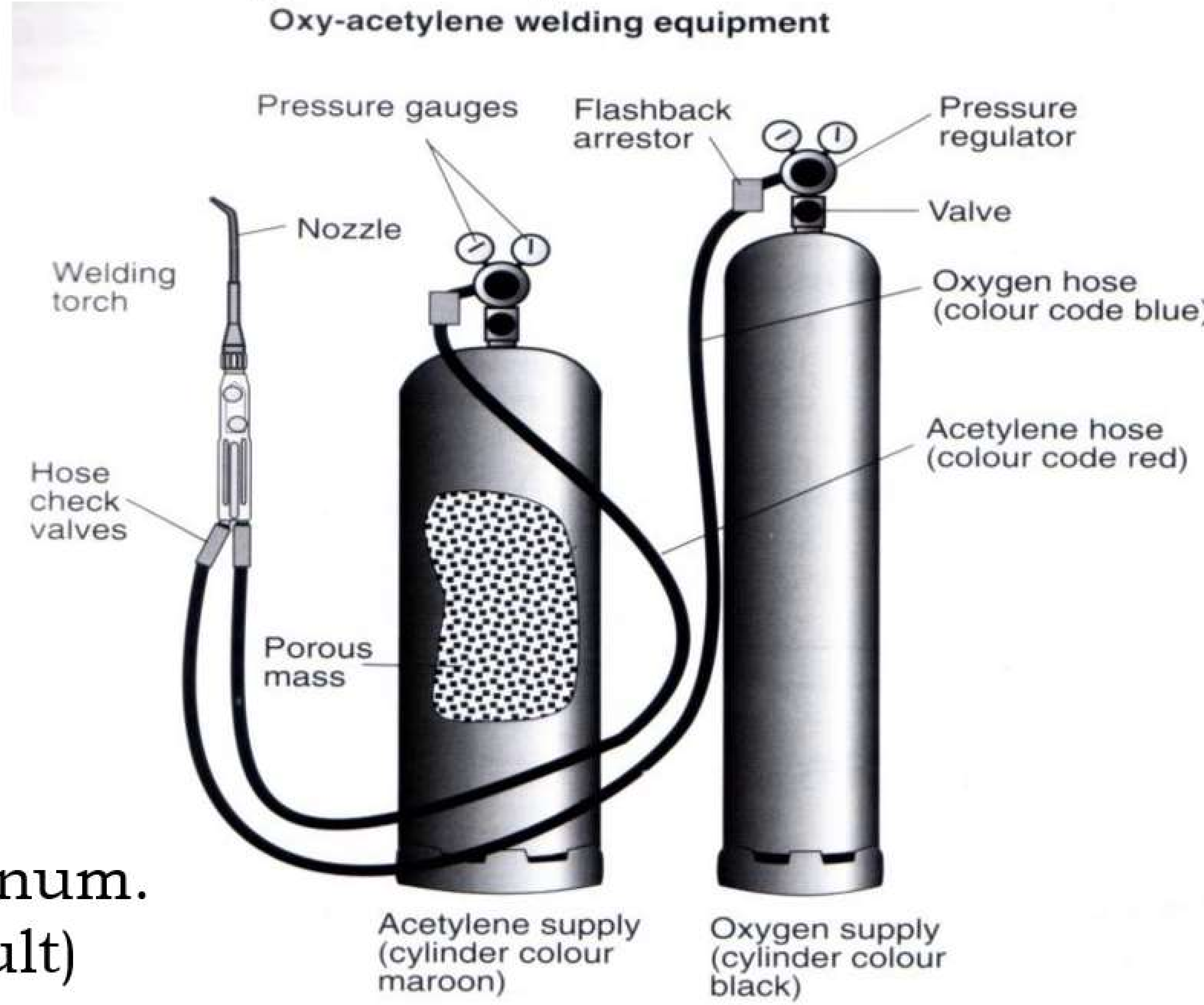
- Types of gas welding flame
  - Oxidizing, reducing , neutral
  - Ratio of acetylene and oxygen
- $C_2H_2+2.5O_2 \rightarrow 2CO_2+H_2O$

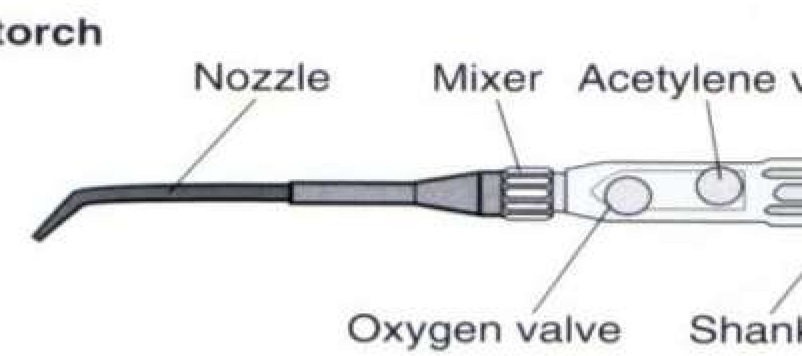
$$C_2H_2 + O_2 \longrightarrow 2CO + H_2 + Heat$$

$$2CO + O_2 \longrightarrow 2CO_2 + Heat$$

$$H_2 + \frac{1}{2}O_2 \longrightarrow H_2O + Heat$$

- Reducing flames used
  - Easily oxidized alloys aluminum.
  - Medium carbon steel (difficult)
- Oxidizing flame is used
  - Welding of brass or bronze
  - Else zinc and tin vaporize

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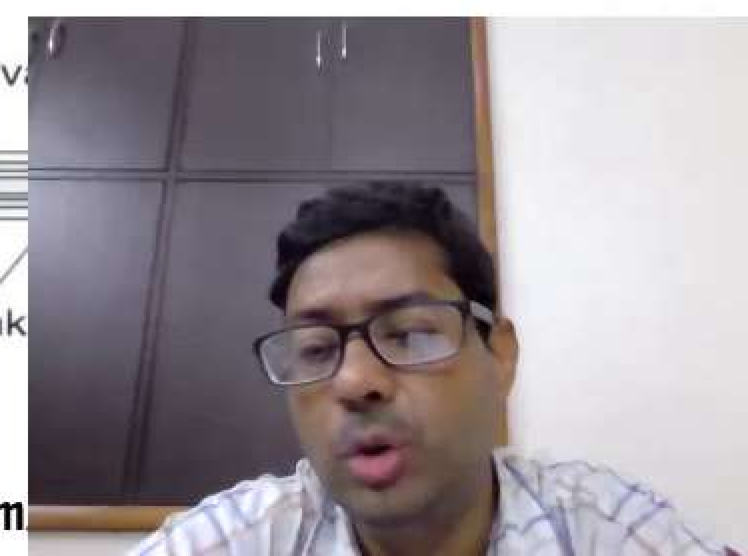
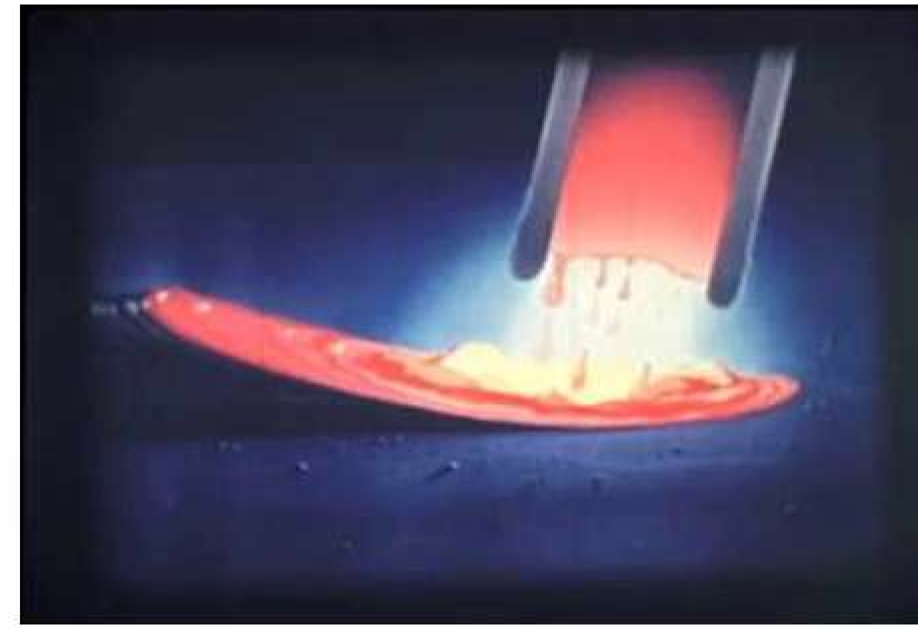


# Arc welding



- Source of heat is an arc
  - Arc is an ionized air (or gas)
    - Source of intense heat
  - Arc is maintained between an electrode and the work piece.
    - Electrode; consumable, non-consumable
  - Typical gap between the electrode and workpiece is 4-5 mm
- Low voltage and high current

https://blog.perfectwelding.fronius.

# Arc initiation in welding

- Arc initiation methods
  - Touch and draw method
    - Touch electrode to produce short circuit & withdraw
    - Metal fumes form (charged ions); rush to negative terminal
    - In the process these ions collide with air molecules
      - Causing these molecules to ionize (+charged and electrons)
    - Charged particles further rush towards opposite terminal
    - In short time an avalanche results and an stable arc forms
  - High voltage and high frequency starting unit
    - High voltage at high frequency isn’t lethal; for short time only
  - Other method include placing steel wo
    electrode & withdrawing

# Basic requirement for arc welding

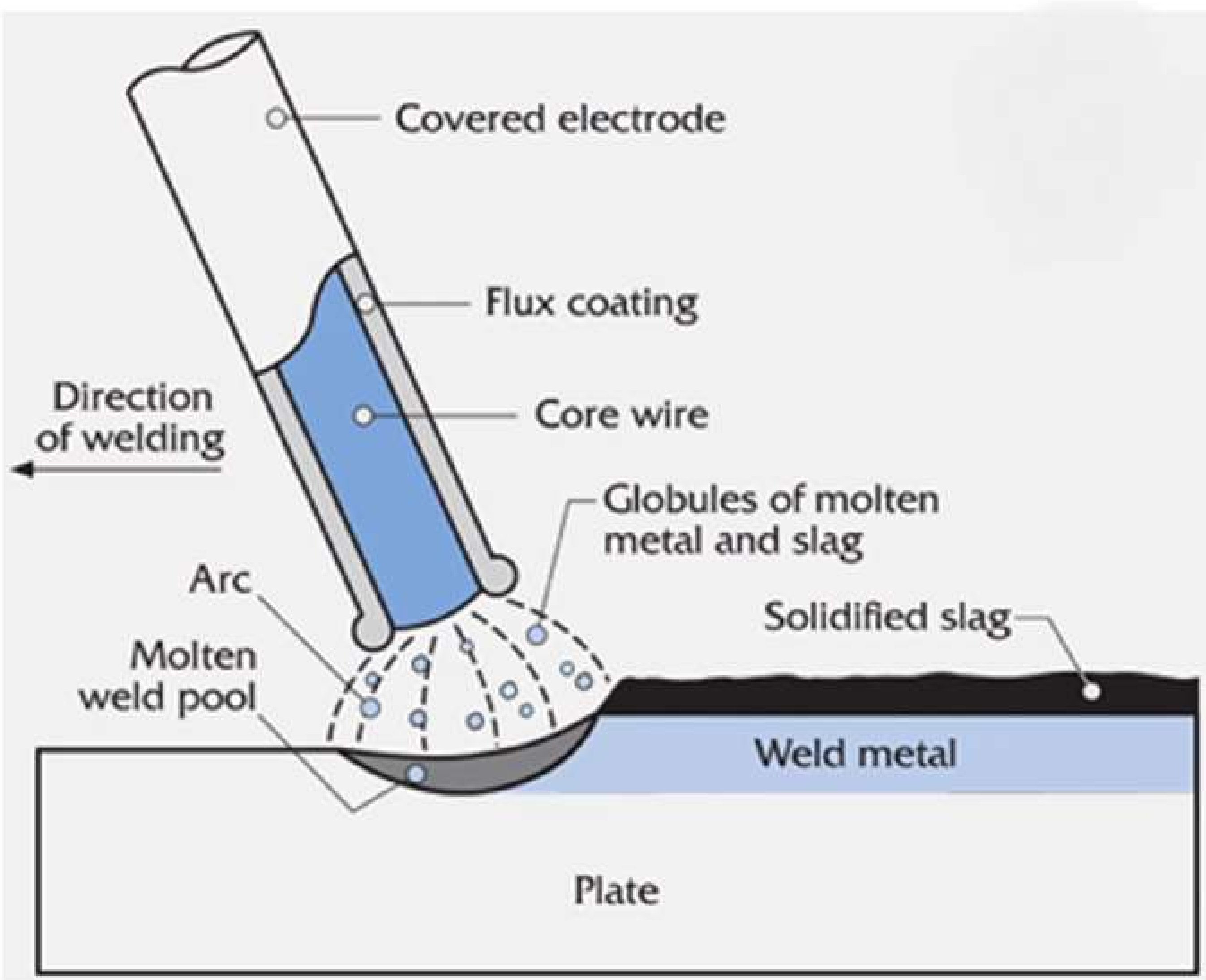
- A power source (AC or DC)
  - Low voltage and high current
    - Motor Generator set, Transformer, Inverter based power sources
  - Should be capable of giving a range of current and voltage
- Electrode and electrode holder
    - consumable or non-consumable
- Earth clamps to connect the workpiece
- Connecting cables
- Shielding of molten metal from atmospheric gases
  - Gas shielding or flux shielding may b

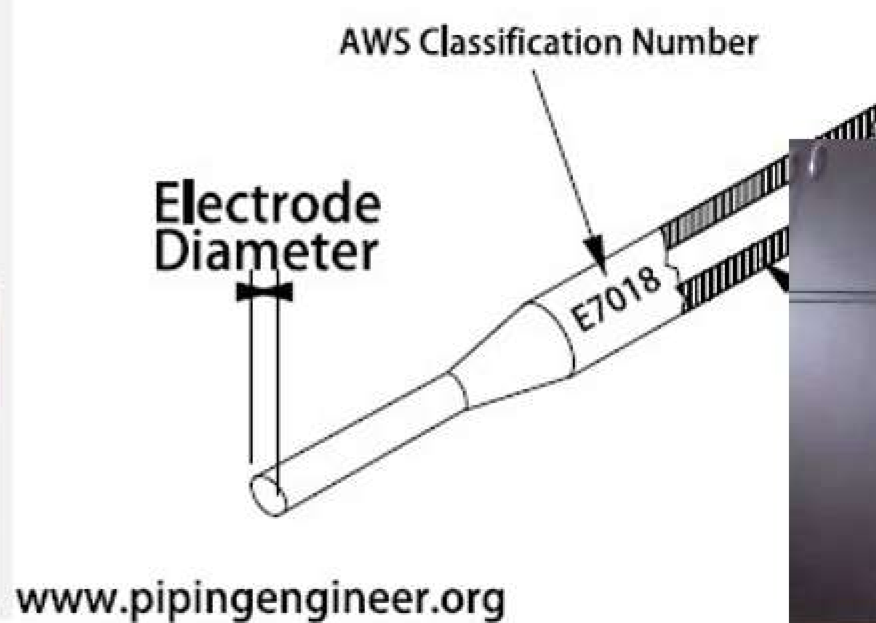
# Shielded Metal Arc welding (SMAW)

- Consumable electrode of 450 mm length
  - Diameter may vary from 2 mm to 5 mm
- A thick layer of flux is applied on electrode
  - Flux also melts along with the electrode in arc
  - The flux provides shielding by following methods
    - Flux melts & covers liquid drop and solidifying metal
    - Generates fumes on heating which displaces air
    - Chemically reacts with any oxide formed
  - Other benefits of flux shielding
    - Offers a means of adding any alloying elements
    - Slows down the cooling rate after welding
- The welding process based on thi manual metal arc welding (MMAV or shielded metal arc welding pro (SMAW)

# Shielded Metal Arc welding (SMAW)

- Flux can be various types
  - Constituent of flux are $CaCO_3$ $CaO$, $Al_2O_3$, $TiO_2$, $Na_2O$,$K_2O$, Cellulose, $CaF_2$, $SiO_2$, Slicate Binder
  - Based on composition Acidic flux, Basic flux, neutral flux
  - AWS coding EXXYZ
    - XX Strength in Psi, Y position Z type of flux & some idea about current etc.



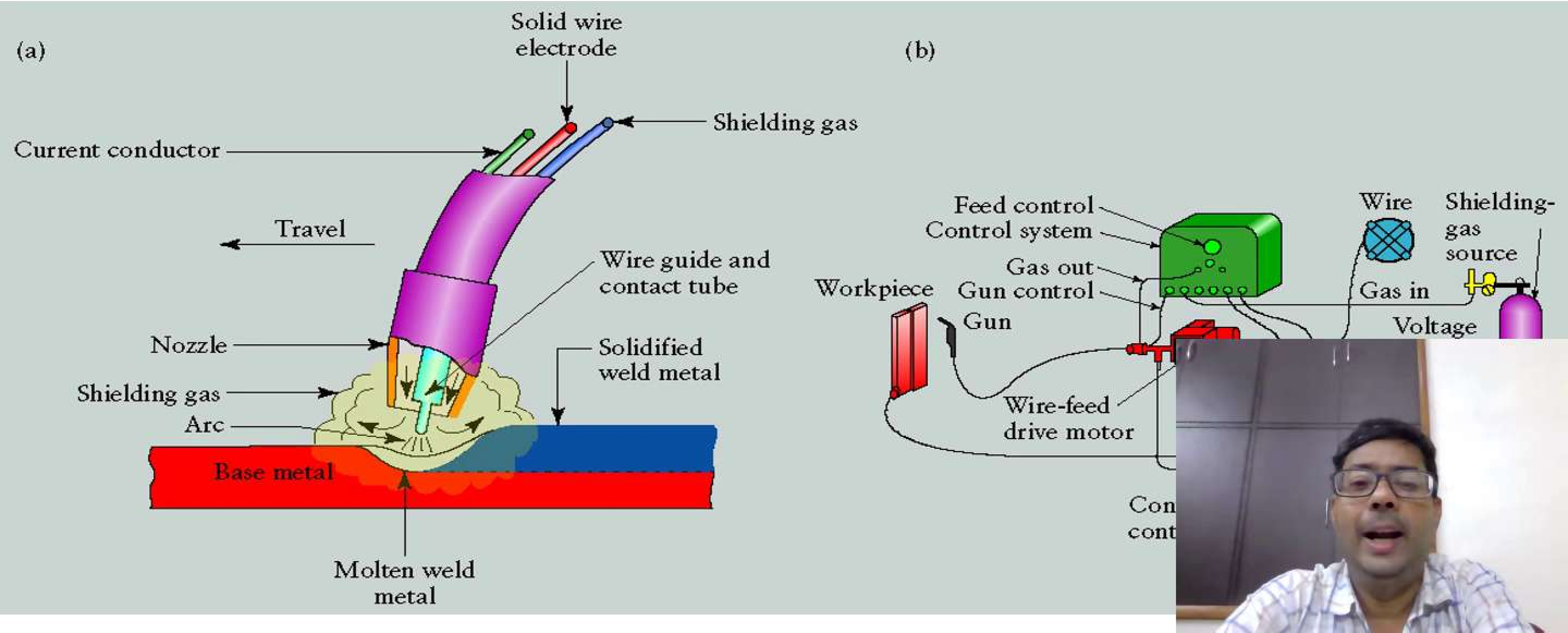




# Issues Shielded Metal Arc welding (SMAW)

- Arc initiation is an issue
- The electrodes have finite length and get consumed
  - Requiring stopping of welding process to change electrode
    - Causes loss in productivity
    - Extinguishing arc and initiation of arc causes defect

# Gas Metal Arc welding (GMAW)

- Filler wire/electrode: bare metal wire in form of a spool
- Shielding by flooding arc with a inter/inactive gas ($CO_2$, Ar, He, $N_2$ and mixture)
  - Diameters commonly available are 0.6mm, 0.8mm, 0.9mm &1.2mm
  - Composition depends on the material being welded
  - For different  electrode, different shielding gas is used



# GMAW setup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